

- 4 A particle X with kinetic energy E_k collides with a stationary particle Y. Both particles have the same mass.

After colliding, X and Y travel together as a single particle.

How much kinetic energy is lost in the collision?

- A zero B $\frac{E_k}{4}$ C $\frac{E_k}{2}$ D $\frac{3E_k}{4}$

Ans: C

$$E_k = \frac{1}{2}mv^2$$

By principle of conservation of momentum,

$$mv + 0 = 2mv_f \rightarrow v_f = \frac{v}{2}$$

$$\text{loss in KE} = KE_i - KE_f$$

$$= \frac{1}{2}mv^2 - \frac{1}{2}(2m)\left(\frac{v}{2}\right)^2 = \frac{1}{2}mv^2 - m\left(\frac{v^2}{4}\right) = \frac{1}{4}mv^2 = \frac{E_k}{2}$$

5 A sphere of mass 3.0 kg travelling due North at 2.0 m s^{-1} collides with another sphere of mass